Notes: Li (RES, 2018)

Better Lucky Than Rich? Welfare Analysis of Automobile Licence Allocations in Beijing and Shanghai

Contents

1 Big picture

- **Question.** How do lottery and auction mechanisms compare when a city allocates scarce vehicle licences, and when vehicle use creates congestion and pollution externalities?
- **Setting.** Beijing allocates new vehicle licences through non-transferable lotteries. Shanghai allocates licences through monthly auctions. Both policies ration the right to register a car, but the allocation rule is very different.
- **Core tradeoff.** Auctions allocate licences to high willingness-to-pay households, which improves allocative efficiency. Lotteries allocate randomly, which creates misallocation, but lotteries may also reduce externalities if high-WTP households drive more and buy less fuel-efficient vehicles.
- **Why the answer is empirical.** If external costs rise sharply enough with willingness to pay, the lottery can dominate the auction. If consumer heterogeneity in WTP is large and the externality gradient is moderate, the auction dominates.
- **Contribution.** The paper estimates the full WTP schedule for vehicle licences and combines it with external cost estimates to compare welfare under alternative allocation mechanisms.
- **Method.** The paper estimates a random-coefficients discrete-choice model of automobile demand in four Chinese cities from 2008 to 2012: Beijing, Nanjing, Shanghai, and Tianjin.
- **Identification strategy.** The paper uses model fixed effects, time effects, city-segment fixed effects, Shanghai-specific time effects, common-trend moments, income-share micro-moments, and quota-matching moments. Shanghai's auction bids create variation in the effective vehicle price, defined as car price plus licence bid.
- **Main sales result.** The quota policies sharply reduce vehicle sales. In the benchmark model, Beijing's lottery reduced new vehicle sales by roughly 60% in 2011 and 59% in 2012. Shanghai's auction reduced sales by roughly 50–70% over 2008–2012.
- **Main welfare result.** In Beijing in 2012, a uniform-price auction would have generated realized consumer surplus of 37.5 billion Yuan, while the lottery realizes only 4.31 billion Yuan. The implied allocative loss from random allocation is 33.19 billion Yuan.
- **Externality result.** The lottery does reduce external costs relative to the auction: for a 15-year horizon, estimated external costs are 29.97 billion Yuan under the lottery and 32.88 billion Yuan under the auction. This advantage is about 2.91 billion Yuan.
- **Net result.** The externality advantage of the lottery is much smaller than the allocative loss. Net social welfare in Beijing in 2012 is estimated at -25.66 billion Yuan under the lottery and 4.62 billion Yuan under a uniform-price auction.
- **Revenue result.** A uniform-price auction would have generated about 19.72 billion Yuan in government revenue in Beijing in 2012, enough to more than cover local public transit subsidies.
- **Big takeaway.** When WTP heterogeneity is large, the allocation rule matters enormously. A lottery may look equitable ex ante, but it can destroy large amounts of surplus by allocating scarce permits to low-WTP households while excluding high-WTP households.

2 Data and measurement (key objects)

2.1 Policy setting: vehicle licence quotas

The policy object is a licence or permit required to register a vehicle. Beijing and Shanghai both restrict the number of new vehicle licences, but they allocate the scarce licences differently.

- **Beijing lottery.** Beijing introduced a quota system in January 2011 after announcing the policy in December 2010. About 20,000 licences per month were allocated by lottery during 2011–2013. The licences are non-transferable.
- **Shanghai auction.** Shanghai has long used auctions for vehicle licences. During 2008–2012, the auction was a monthly online, multi-unit, discriminatory, dynamic auction. Bidders first submit an initial bid and later can revise bids within a narrow window around the current lowest accepted bid.
- **Quota and treatment intensity.** Beijing's rationing appears as a lottery probability. Shanghai's rationing appears as a monetary licence price, the winning bid.

Interpretation. The policies provide a rare comparison of two allocation rules for the same broad policy objective. Beijing gives away a scarce good randomly; Shanghai prices it.

2.2 Cities used in the empirical design

The paper studies four cities:

- Beijing: lottery treatment city after 2010.
- Shanghai: auction city throughout the sample.
- Tianjin: nearby comparison city for Beijing.
- Nanjing: nearby comparison city for Shanghai.

Interpretation. Nanjing and Tianjin do not have licence quotas during the sample, so they help discipline the counterfactual path of vehicle demand absent the policies.

2.3 Vehicle sales data

The main market-level dataset contains monthly vehicle sales by model and city from 2008 to 2012. A model is a vintage-nameplate. The data contain 84,912 city-model-month observations and 1,769 distinct models.

Interpretation. This is the core demand dataset. It tells the econometrician how sales vary across models, months, cities, and policy regimes.

2.4 Vehicle characteristics

For each vehicle model, the paper observes characteristics including:

- vehicle price;
- fuel economy, measured in litres per 100 km;
- vehicle size;
- engine size;
- vehicle type and segment.

Interpretation. These characteristics enter the discrete-choice demand model. They also connect vehicle choices to externalities because larger, less fuel-efficient vehicles generate more gasoline-related external costs.

2.5 Income distribution and buyer-income microdata

The paper constructs household income distributions for each city-year using city statistical yearbooks and the Chinese Household Income Survey. It also uses Ford Motor Company survey information on the income-group shares of new vehicle buyers by city and year.

Interpretation. These buyer-income shares are crucial. They reveal that high-income households are overrepresented among new car buyers, which pins down the relationship between income, price sensitivity, and WTP.

2.6 Externality proxy from travel survey data

The paper uses the 2010 Beijing Household Travel Survey to show that higher-income households own larger-engine vehicles and drive more. Table 1 shows, for example, that annual VMT rises from about 13,907 km in the lowest income group to about 20,382 km in the highest income group.

Interpretation. This motivates why external costs are type-dependent. If high-WTP households tend to be high-income households, then the auction selects households that generate more vehicle-use externalities.

2.7 External cost calibration

The welfare analysis uses external cost estimates from prior work on Beijing automobile use. The preferred value corresponds to externalities from congestion, local air pollution, traffic accidents, and carbon emissions. The gross external cost is converted into a per-litre gasoline externality; after accounting for China's gasoline tax, the paper uses an uninternalized external cost measure.

Interpretation. The externality component is not estimated from the demand model. It is brought in from the transportation and environmental economics literature and combined with simulated vehicle choices.

2.8 Key measured objects

- **WTP for a licence.** The consumer surplus from owning the most preferred vehicle, interpreted as the extra amount a household would pay for a licence.

- **Realized consumer surplus.** Total surplus among households that actually receive licences under a mechanism.
- **Unrealized consumer surplus.** Surplus lost when high-WTP households do not receive a licence.
- **External costs.** Discounted congestion and pollution-related costs from vehicle use.
- **Net social welfare.** Realized consumer surplus minus external costs. Auction revenue is reported separately because it is a transfer from households to government.

3 Models and empirical specifications

3.1 Simple welfare logic

Let households be indexed by willingness to pay for a licence, v_i, and let the external cost generated by household i be e_i. For a quota Q, an efficient private-value auction gives licences to the Q households with highest v_i. A lottery gives licences randomly.

A simple comparison is

$$\Delta W_{A-L} = \underbrace{\left[\sum_{i \in A(Q)} v_i - \sum_{i \in L(Q)} v_i \right]}_{\text{allocative gain from auction}} - \underbrace{\left[\sum_{i \in A(Q)} e_i - \sum_{i \in L(Q)} e_i \right]}_{\text{extra external costs under auction}}, \tag{1}$$

where $A(Q)$ is the set allocated licences by the auction and $L(Q)$ is the set allocated licences by the lottery.

Interpretation. The auction dominates when the allocative gain is larger than the additional external costs it creates. The lottery can dominate only if high-WTP winners create sufficiently larger external costs.

3.2 Individual utility

Household i's indirect utility from vehicle model j in market m and month t is

$$u_{mtij} = \bar{u}(p_j, b_{mti}, X_j, \xi_{mtj}, y_{mti}, Z_{mti}) + \varepsilon_{mtij}, \tag{2}$$

where p_j is the vehicle price, b_{mti} is the licence bid or licence payment, X_j are vehicle characteristics, ξ_{mtj} are unobserved product attributes and demand shocks, y_{mti} is household income, and Z_{mti} are unobserved household demographics.

Interpretation. The consumer chooses among car models and the outside option. The licence policy changes effective purchase costs through either lottery access or auction bid payments.

3.3 Deterministic utility

The deterministic part of utility is specified as

$$\bar{u}_{mtij} = \alpha_{mti} \ln(p_j + b_{mti}) + \sum_{k=1}^{K} X_{mtjk} \widetilde{\beta}_{mtik} + \xi_{mtj}. \tag{3}$$

The price coefficient is household-specific:

$$\alpha_{mti} = \alpha_0 + \alpha_1 \ln y_{mti} + \sigma \nu_{mti}. \tag{4}$$

Preferences over non-price characteristics are also random:

$$\widetilde{\beta}_{mtik} = \bar{\beta}_k + \sigma_k^u \nu_{mtik}. \tag{5}$$

Putting these pieces together,

$$u_{mtij} = (\alpha_0 + \alpha_1 \ln y_{mti} + \sigma \nu_{mti}) \ln(p_j + b_{mti}) + \sum_{k=1}^{K} X_{mtjk}(\bar{\beta}_k + \sigma_k^u \nu_{mtik}) + \xi_{mtj} + \varepsilon_{mtij}. \tag{6}$$

Interpretation. The model allows rich heterogeneity. Income affects price sensitivity, and unobserved household tastes affect preferences for price and characteristics.

3.4 Choice probabilities

With type-I extreme value idiosyncratic shocks, the conditional choice probability is

$$Pr_{mtij} = \frac{\exp(\bar{u}_{mtij})}{1 + \sum_{h=1}^{J} \exp(\bar{u}_{mtih})}. \tag{7}$$

Interpretation. This is the logit probability conditional on household characteristics and random coefficients. The outside option is normalized into the denominator through the “1” term.

3.5 Market shares without a quota

In cities and periods without a quota, aggregate market share for product j is

$$S_{mtj} = \int Pr_{mtij}(p_j, X_j, \xi_{mtj}, y_{mti}, Z_{mti})\, dF(y_{mti}, Z_{mti}). \tag{8}$$

Interpretation. Market demand averages individual choice probabilities over the simulated distribution of household incomes and tastes.

3.6 Probability that a buyer needs a new licence

Some buyers replace an existing vehicle and do not need a new licence. The paper models the probability that a buyer needs a new licence as

$$L_{mt} = \frac{1}{1 + \exp(\gamma_0 + o_{mt}\gamma_1)}, \tag{9}$$

where o_{mt} is the vehicle ownership rate.

Interpretation. As vehicle ownership rises, a larger share of purchases are replacement purchases, so fewer buyers need new licences.

3.7 Beijing lottery market share

For Beijing after the policy starts, households who need a new licence obtain one only with lottery probability ρ. The model writes the market share as

$$S_{1tj} = \int [1(c_{1t} = 1)Pr_{1tij}(\cdot)\rho + 1(c_{1t} = 0)Pr_{1tij}(\cdot)]\, dF(y_{1ti}, Z_{1ti}, c_{1t}). \tag{10}$$

Interpretation. If a household needs a new licence, its purchase probability is scaled by the lottery success probability. If it does not need a new licence, it can buy without this lottery constraint.

3.8 Shanghai auction market share

For Shanghai, households who need a new licence pay the auction bid; replacement buyers do not. The market share is

$$S_{3tj} = \int [1(c_{3t} = 1)Pr_{3tij}(b_{mti} > 0, \cdot) + 1(c_{3t} = 0)Pr_{3tij}(b_{mti} = 0, \cdot)]\, dF(y_{3ti}, Z_{3ti}, c_{3t}). \tag{11}$$

Interpretation. Shanghai's policy raises the effective purchase price for households that need a licence. This creates identifying variation in demand responses to price plus bid.

3.9 BLP mean utility and contraction mapping

The utility can be decomposed into mean utility, household-specific utility, and idiosyncratic taste:

$$u_{mtij} = \delta_{mtj} + \mu_{mtij} + \varepsilon_{mtij}. \tag{12}$$

Mean utility is modeled as

$$\delta_{mtj} = \delta_j + \eta_t + 1(m = 3)\eta_t' + \zeta_{ms} + \kappa_m yr_t + e_{mtj}, \tag{13}$$

where δ_j are model fixed effects, η_t are time effects, η'_t are Shanghai-specific time effects, ζ_{ms} are city-segment effects, and $\kappa_m yr_t$ are city-specific trends.

For given nonlinear parameters, the BLP contraction recovers δ:

$$\delta_{mt}^{n+1} = \delta_{mt}^{n} + \ln(S_{mt}^{o}) - \ln\left[\widehat{S}(\delta_{mt}^{n}, \theta_2, \theta_3)\right]. \tag{14}$$

Interpretation. The model chooses mean utilities that make predicted product shares match observed product shares. Then the remaining structure identifies the parameters governing heterogeneity and policy constraints.

3.10 Moment conditions

The GMM estimator stacks three groups of moments.

First, city-year demand shocks have zero mean conditional on city-year indicators:

$$E\left[e_{mtj}(\theta_2, \theta_3) \mid d_{mt}\right] = 0. \tag{15}$$

Second, predicted income-group shares among vehicle buyers match observed buyer-income shares:

$$E_t\left[\widehat{S}_{mgt|buyers}(\theta_2, \theta_3) - S_{mgt|buyers}\right] = 0, \tag{16}$$

where

$$\widehat{S}_{mgt|buyers}(\theta_2, \theta_3) = \frac{\sum_{i=1}^{N_{mt}} d(y_{mti} \in INC_g) \sum_{j=1}^{J} Pr_{mtij}}{\sum_{i=1}^{N_{mt}} \sum_{j=1}^{J} Pr_{mtij}}. \tag{17}$$

Third, predicted licence quantities match observed quotas:

$$E_t\left[\widehat{Q}_{mt}(\theta_2, \theta_3) - Q_{mt}\right] = 0. \tag{18}$$

Interpretation. The first moments discipline counterfactual policy effects, the second moments discipline heterogeneity in WTP, and the third moments discipline licence-use probabilities and the lottery stringency parameter.

3.11 WTP for a licence

The paper defines household i's WTP for a licence as expected consumer surplus from the most preferred vehicle:

$$E(CS_i) = E\left[\max_{j=1,\ldots,J} \frac{\bar{u}_{ij} + \varepsilon_{ij}}{MU_{ij}}\right]. \tag{19}$$

If marginal utility of money were constant, this would simplify to the log-sum expression

$$E(CS_i) = \frac{1}{MU} \ln \left(\sum_{j=1}^{J} \exp(\bar{u}_{ij}) \right). \tag{20}$$

But because the utility specification is nonlinear in price, the paper uses simulation methods rather than relying on this closed-form expression.

Interpretation. The WTP curve is a distribution of households' consumer surplus from vehicle ownership. This is the central object for comparing lottery and auction allocations.

3.12 Welfare calculation

For mechanism $r \in \{L, A\}$, where L is lottery and A is auction, net social welfare is

$$NSW_r = CS_r - EC_r. \tag{21}$$

The allocative cost of lottery relative to auction is

$$AC_L = CS_A - CS_L. \tag{22}$$

The externality advantage of lottery is

$$EA_L = EC_A - EC_L. \tag{23}$$

The welfare gain from switching from lottery to auction is

$$NSW_A - NSW_L = (CS_A - CS_L) - (EC_A - EC_L) = AC_L - EA_L. \tag{24}$$

Interpretation. Lottery is better only if its externality advantage exceeds its allocative cost. In the estimates, AC_L is about 33.19 billion Yuan while EA_L is only about 2.91 billion Yuan under the 15-year horizon.

4 Identification and instruments

4.1 What identifies the price and WTP parameters?

The central challenge is that WTP is not directly observed. Beijing has no price for licences because licences are allocated by lottery, and resale is not allowed. Shanghai has licence bids, but the auction format is non-standard and bids may not reveal true values.

The paper therefore estimates WTP indirectly from vehicle demand. A licence is valuable because it allows a household to buy and use a vehicle. Consumer surplus from the best vehicle option is used as the household's WTP for the licence.

Interpretation. The paper turns the licence valuation problem into a vehicle demand estimation problem.

4.2 Model fixed effects and product quality

A standard BLP concern is that vehicle prices are correlated with unobserved product quality. The paper's multi-city setting allows it to include vehicle model fixed effects. These absorb product-level attributes that are constant across cities and time.

Interpretation. This avoids relying only on the classic BLP assumption that observed and unobserved product characteristics are mean independent.

4.3 Shanghai bid variation

In Shanghai, some buyers need a new licence and pay the auction bid; replacement buyers do not. Therefore, for the same vehicle model, effective purchase costs differ depending on whether the buyer must pay the bid.

Interpretation. This policy-induced variation helps identify price sensitivity because the bid enters household-specific utility, not mean utility. The model can control for model fixed effects while still using variation in effective prices.

4.4 Common-trend moments

The first set of moments imposes a common-trend assumption across cities after controlling for time effects, city-segment effects, Shanghai-specific time effects, and city trends. Beijing's post-2010 deviation is interpreted as the effect of the lottery policy rather than an unrelated negative demand shock.

Interpretation. This is similar in spirit to a difference-in-differences design. Figure 2 and Table 3 support the common-trend assumption by showing that pre-policy trends are broadly similar.

4.5 Micro-moments from buyer income shares

The buyer-income micro-moments match the shares of new vehicle buyers from different income groups. High-income households account for a disproportionately large share of new vehicle buyers.

Interpretation. These moments are essential for the curvature of the WTP schedule. Without them, the model can fit sales but produce implausible heterogeneity and price sensitivity patterns.

4.6 Quota moments

The third moment group matches predicted licence quantities to observed quotas. This disciplines ρ, the lottery stringency parameter, and the probability that a buyer needs a new licence.

Interpretation. Sales data alone do not fully reveal whether low purchases reflect weak demand or rationing. Quota moments help separate demand from allocation constraints.

4.7 Reduced-form support

The paper estimates pre-policy trend regressions and difference-in-differences style sales regressions. The reduced-form evidence shows that Beijing, Nanjing, and Tianjin track each other before Beijing's lottery policy, and that the policy sharply reduces Beijing sales after implementation.

Interpretation. The reduced-form results are not the final welfare model, but they validate the identifying variation used by the structural model.

4.8 Remaining limitations

- **No randomized mechanism assignment.** Beijing and Shanghai choose different mechanisms for institutional and historical reasons, so the city-level comparison is not randomized.
- **Static demand.** The model is static even though cars are durable goods and households may wait strategically.
- **WTP horizon.** The WTP interpretation assumes a licence lasts roughly one vehicle lifetime. If households value licences over multiple vehicle lifetimes, the estimates are conservative.
- **Externality calibration.** External costs are taken from prior estimates and depend on assumptions about vehicle use, fuel economy, and time horizon.
- **Auction benchmark.** The welfare comparison uses a uniform-price auction as the efficient allocation benchmark, not necessarily the exact non-standard Shanghai auction format.

Interpretation. The evidence is unusually rich for a welfare comparison of allocation mechanisms, but the conclusions still rely on structural assumptions and calibrated external costs.

5 Tables and figures

5.1 Figure 1: Welfare comparison of lottery and auction with externalities

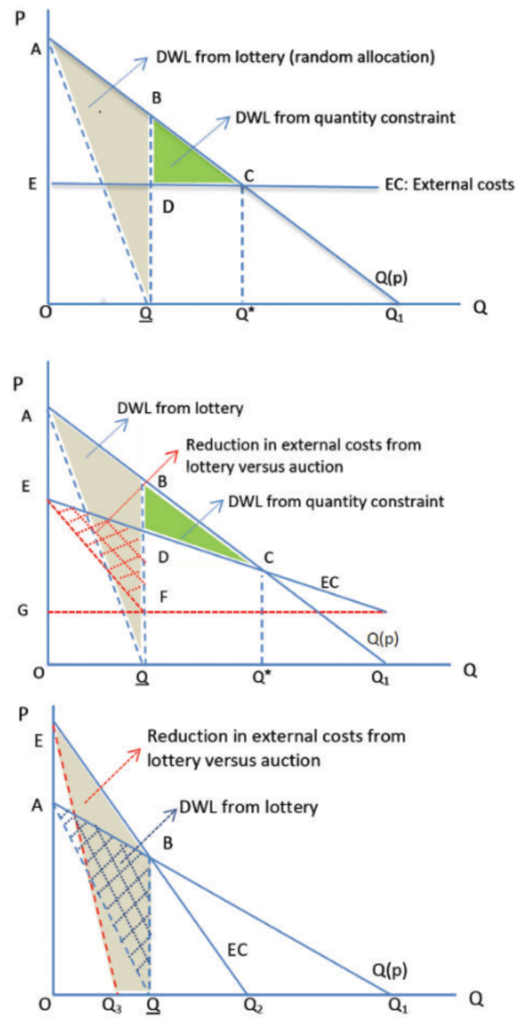

FIGURE 1
Welfare comparison of lottery and auction with externalities.

Figure 1: theoretical welfare comparison of lottery and auction.

Read: The figure illustrates three cases. With constant external costs, the auction dominates because it allocates licences to the highest-WTP households. With external costs increasing in WTP, the lottery gains an externality advantage. If external costs rise extremely steeply with WTP, the lottery can dominate.

Economic message. The paper's empirical task is to determine which case describes Beijing. The estimates show that Beijing looks like the middle case: the lottery lowers externalities, but not enough to offset misallocation.

5.2 Table 1: Income, engine size, and travel

TABLE 1

Household income, engine size and annual travel

Annual income (Yuan)	Engine size(litre) Mean	S.D.	Annual VMT(km) Mean	S.D.	Number of observations
≤50k	1.58	0.38	13,907	8,499	4,827
50–100k	1.68	0.38	14,922	8,813	5,379
100–150k	1.78	0.42	16,554	9,478	1,327
150–200k	1.83	0.43	17,882	9,937	417
200–250k	1.87	0.39	17,661	9,348	133
250–300k	1.94	0.48	18,923	9,328	70
≥ 300k	2.18	0.63	20,382	12,480	98
Full sample	1.67	0.40	14,896	8,927	12,251

Notes: The data source is 2010 Beijing Household Travel Survey. There are 12,251 vehicles surveyed from 10,577 households. In all, 1,440 households own two vehicles while 93 households own three vehicles.

Table 1: higher-income households drive more and own larger-engine vehicles.

Read: Engine size rises from 1.58 litres in the lowest income group to 2.18 litres in the highest income group. Annual VMT rises from about 13,907 km to about 20,382 km.

Economic message. This table gives the empirical motivation for type-dependent externalities. The auction selects higher-WTP households, and these households are likely to impose larger congestion and pollution costs.

5.3 Figure 2: New vehicle sales trends in four cities

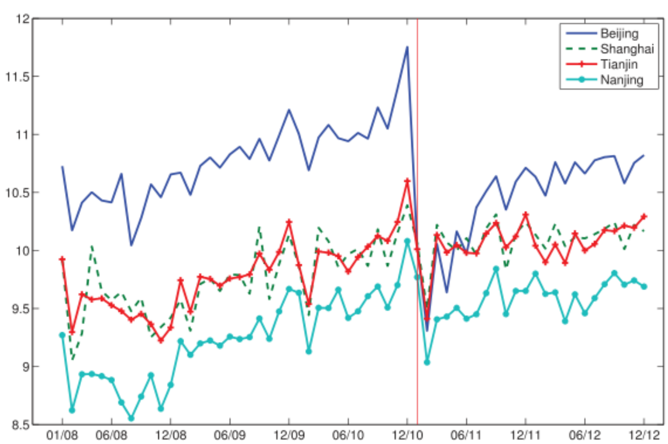

Figure 2: monthly new vehicle sales in Beijing, Shanghai, Tianjin, and Nanjing.

Read: Before the Beijing policy, sales in Beijing, Tianjin, and Nanjing move together. After the policy announcement, Beijing sales sharply fall relative to the controls.

Economic message. The graph supports the common-trend assumption behind the structural moments and the difference-in-differences logic. It also shows the announcement effect around December 2010.

5.4 Table 4: GMM demand estimates

TABLE 4
Parameter estimates from GMM

Variables	Specification 1		Specification 2		Specification 3		Specification 4		Specification 5	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
Parameters in the household-specific utility (θ_2)										
Ln(price+bid)	−16.810	0.225	−17.080	0.200	−14.515	0.185	−17.323	0.173	−16.571	0.173
Ln(price+bid)*Ln(income)	1.620	0.033	1.888	0.031	1.443	0.054	1.620	0.026	1.620	0.037
Ln(income)	−6.394	0.366	−7.619	0.352	−5.807	0.409	−6.404	0.301	−6.418	0.287
σ for Ln(price+bid)	0.067	0.014	0.073	0.026	0.073	0.016	0.069	0.012	0.067	0.027
σ for constant	0.655	0.133	0.698	0.415	0.714	0.152	0.583	0.184	0.643	0.304
σ for engine displacement	0.645	0.289	0.708	0.041	0.708	0.250	0.671	0.276	0.648	0.347
Auxiliary parameters for allocation mechanism (θ_3)										
ρ	0.229	0.030	0.384	0.087	0.236	0.039	0.220	0.034	0.231	0.030
γ_0	−1.426	0.203	−1.111	0.406	−1.422	0.262	−1.428	0.185	−1.429	0.179
γ_1	0.259	0.020	0.139	0.061	0.250	0.026	0.259	0.035	0.260	0.038

Table 4: parameter estimates from the random-coefficients demand model.

Read: The benchmark specification has a strongly negative coefficient on $\ln(price + bid)$ and a positive interaction between $\ln(price + bid)$ and $\ln(income)$. The lottery stringency parameter ρ is about 0.229.

Economic message. Higher-income households are less price sensitive. The estimated ρ implies severe rationing: only about one out of five potential buyers needing a licence obtain one through the lottery in the model.

5.5 Table 5: Policy impacts on sales

TABLE 5
Policy impacts on sales in Beijing and Shanghai

Year	Observed	Counterfactual sales without licence quotas in Beijing and Shanghai				
	Sales	Specification 1	Specification 2	Specification 3	Specification 4	Specification 5
Beijing						
2011	334,308	833,431	612,655	843,796	848,434	830,453
2012	520,442	1,274,954	940,851	1,289,578	1,297,333	1,269,863
Shanghai						
2008	165,298	373,569	319,957	356,631	385,944	369,692
2009	208,570	482,096	413,194	461,496	498,211	478,276
2010	264,232	662,500	567,299	635,184	684,922	659,805
2011	277,119	744,558	629,219	715,686	769,095	739,892

Table 5: observed sales and counterfactual sales without licence quotas.

Read: In Beijing, observed sales are 334,308 in 2011 and 520,442 in 2012. Under the benchmark counterfactual without the quota, sales would be 833,431 in 2011 and 1,274,954 in 2012. In Shanghai, observed 2012 sales are 295,047 while benchmark counterfactual sales without the auction policy are 869,915.

Economic message. Both policies are quantitatively important. The quota systems are not small frictions; they cut new vehicle sales by large margins.

5.6 Figure 3: WTP schedules for licences

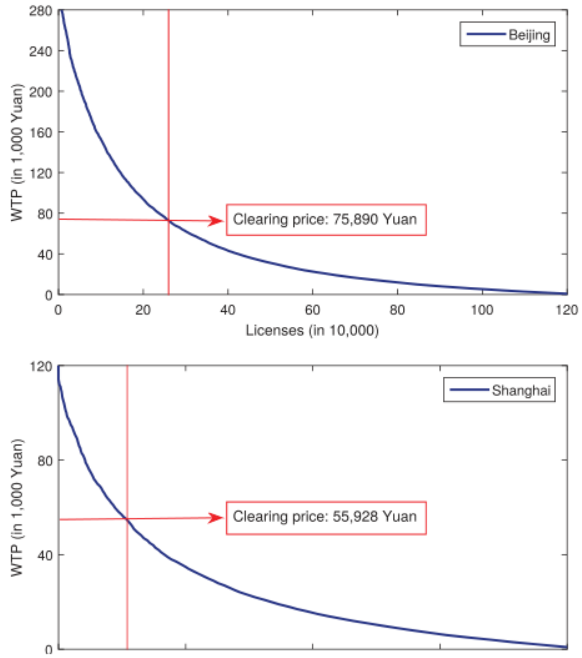

Figure 3: estimated WTP schedules for licences in Beijing and Shanghai.

Read: The WTP curve is steep at the top and has a long right tail. In Beijing, the uniform-price auction clearing price at the observed quota would be 75,890 Yuan. In Shanghai, the comparable clearing price is 55,928 Yuan.

Economic message. The WTP distribution is highly heterogeneous. This is why random allocation is costly: many licences go to relatively low-WTP households while high-WTP households remain rationed.

5.7 Figure 4: WTP and external costs

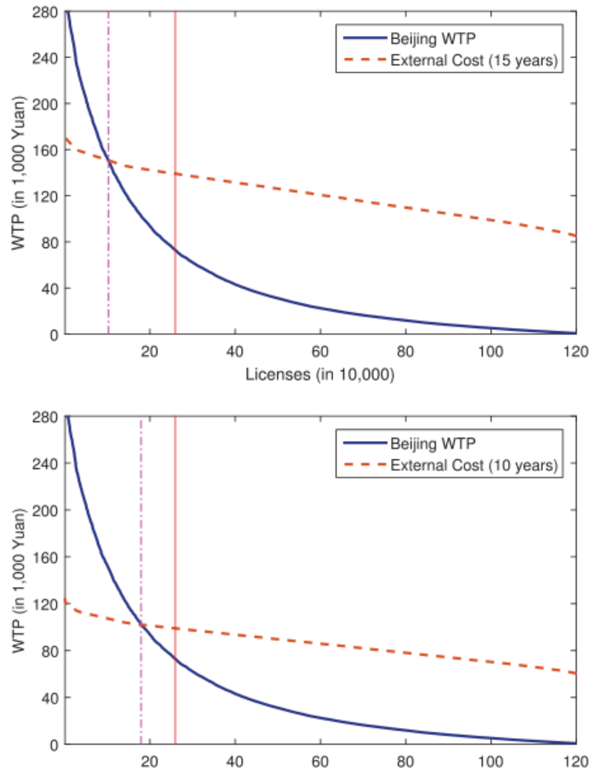

Figure 4: WTP schedule and external cost curves for Beijing.

Read: The dashed external cost curves lie above the WTP curve for many lower-WTP households. The optimal quota is below the observed quota: 111,404 under the 15-year external-cost horizon and 179,407 under the 10-year horizon.

Economic message. The quota level matters separately from the allocation mechanism. The observed Beijing quota is too high relative to the externality-adjusted optimum in the model.

5.8 Table 6: Welfare comparison in Beijing in 2012

TABLE 6

Welfare comparison between lottery and auction in Beijing in 2012

	15-year Horizon		10-year Horizon	
	Lottery (1)	Auction (2)	Lottery (3)	Auction (4)
Observed quota level				
Quota in 2012	259,800	259,800	259,800	259,800
Clearing price (in 1,000 Yuan)	0.00	75.89	0.00	75.89
Realized consumer surplus (in billion)	4.31	37.50	4.31	37.50
Unrealized consumer surplus (in billon)	33.19	0.00	33.19	0.00
Average fuel economy (liters per 100 km)	9.14	9.55	9.14	9.55
Annual VMT (1,000 kms)	16.35	17.75	16.35	17.75
Total external costs (in billion)	29.97	32.88	22.29	24.46
Government revenue (in billion)	0.00	19.72	0.00	19.72
Net social welfare (in billion)	−25.66	4.62	−17.99	13.04
Optimal quota level				
Optimal quota		111,404		179,407
Clearing price (in 1,000)		143.50		102.50
Consumer surplus (in billion)		30.04		40.50
Total external costs (in billion)		14.40		17.04
Government revenue (in billion)		15.99		18.39
Net social welfare		15.64		23.47

Table 6: welfare comparison between lottery and uniform-price auction in Beijing.

Read: At the observed quota, realized consumer surplus is 4.31 billion Yuan under the lottery and 37.50 billion Yuan under the auction. The 15-year external costs are 29.97 billion Yuan under the lottery and 32.88 billion Yuan under the auction. Net social welfare is -25.66 billion Yuan under the lottery and 4.62 billion Yuan under the auction.

Economic message. The lottery's externality advantage is economically meaningful but too small. The allocative loss from random allocation dominates, producing a welfare loss of about 30 billion Yuan relative to auction.

6 Short summary

- Beijing uses a lottery to allocate vehicle licences; Shanghai uses an auction.
- The main welfare tradeoff is allocative efficiency versus type-dependent externalities.
- The paper estimates WTP for licences using a random-coefficients automobile demand model.
- Model identification comes from multi-city sales data, Shanghai bid variation, common-trend restrictions, income-share micro-moments, and quota moments.
- High-income households are more likely to buy cars, buy larger vehicles, and drive more.
- The lottery does reduce externalities relative to auction because it allocates fewer licences to high-WTP, high-driving households.
- But the lottery creates a much larger misallocation cost because licences are randomly assigned.
- In Beijing in 2012, the allocative loss is 33.19 billion Yuan, while the 15-year externality advantage is only 2.91 billion Yuan.
- Net social welfare is about 30 billion Yuan lower under the lottery than under a uniform-price auction.

- Auction revenue would also be large: around 19.72 billion Yuan in Beijing in 2012.

7 Conclusion

Li's paper shows that allocation mechanisms matter not only theoretically but quantitatively. In a setting with scarce vehicle licences, a non-transferable lottery may appear fair because it gives everyone an equal chance, but it does not direct licences to households that value them most.

The key economic insight is that type-dependent externalities make the comparison nontrivial. If high-WTP households impose much higher external costs, a lottery can improve welfare by keeping some high-externality users out of the market. The paper finds that this effect exists in Beijing, but it is far too small to offset the consumer surplus loss from random allocation.

The main policy lesson is that the choice between lottery and auction cannot be evaluated only by equity concerns or by whether the policy reduces vehicle ownership. The allocation rule determines who receives scarce permits, and therefore determines the size of the surplus loss. In Beijing's case, a uniform-price auction would have delivered much higher net welfare and substantial public revenue that could have been used to improve public transit or address distributional concerns.